

WARSZAWA OKECIE, Poland

WMO: 123750

Lat: 52.1628N

Lon: 20.9611E

Elev: 349

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.1	(c) 10.0	(d) -0.8	(e) 5.3	(f) 5.4	(g) 4.5	(h) 7.0	(i) 11.5	(j) 27.4	(k) 40.9	(l) 24.3	(m) 37.1	(n) 6.4	(o) 90	(p) 0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.8	(c) 86.4	(d) 68.6	(e) 83.1	(f) 67.0	(g) 80.2	(h) 65.6	(i) 70.8	(j) 82.3	(k) 69.2	(l) 79.3	(m) 67.5	(n) 76.7	(o) 8.1	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	101.2	74.9	65.7	96.0	73.5	64.2	91.1	72.1	34.9	82.7	33.5	79.3	32.1	76.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.5	(b) 18.9	(c) 16.7	(e) -0.3	(f) 92.1	(g) 6.5	(h) 3.1	(i) -4.9	(j) 94.4	(k) -8.7	(l) 96.2	(m) -12.4	(n) 97.9	(o) -17.1	(p) 100.1		
(4) 21.5	(5) 18.9	(6) 16.7	(7) -0.8	(8) 73.7	(9) 6.4	(10) 1.6	(11) -5.4	(12) 74.9	(13) -9.1	(14) 75.8	(15) -12.7	(16) 76.7	(17) -17.3	(18) 77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	28.6	31.1	37.7	48.5	57.6	63.5	67.5	66.3	57.3	47.9	39.1	31.5		
	DBStd	15.69	9.95	9.07	7.98	7.78	7.13	6.28	5.92	5.73	6.21	7.33	7.61	9.18		
	HDD50	2756	665	530	386	117	15	0	0	0	9	128	332	574		
	HDD65	6454	1130	950	847	495	249	103	41	52	240	530	778	1039		
	CDD50	2077	0	0	4	73	250	404	543	505	229	64	5	0		
	CDD65	297	0	0	0	1	18	57	119	92	10	0	0	0		
	CDH74	2623	0	0	0	32	204	523	1011	760	93	0	0	0		
	CDH80	702	0	0	0	4	32	130	309	211	16	0	0	0		
	WSAvg	8.1	9.1	8.9	9.2	8.3	7.7	7.7	7.2	6.7	7.2	7.9	8.3	8.5		
	PrecAvg	21.1	1.1	1.0	1.1	1.4	2.2	2.6	3.1	2.4	1.7	1.5	1.5	1.3		
(15) Precipitation	PrecMax	31.4	3.1	2.8	2.6	3.7	7.2	5.9	11.6	7.0	5.4	5.8	4.3	3.2		
	PrecMin	13.4	0.0	0.0	0.2	0.2	0.6	0.6	0.6	0.2	0.3	0.1	0.0	0.1		
	PrecStd	3.8	0.6	0.6	0.5	0.7	1.1	1.3	2.0	1.3	1.1	1.2	0.8	0.7		
	PrecStd	3.8	0.6	0.6	0.5	0.7	1.1	1.3	2.0	1.3	1.1	1.2	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	50.2	52.9	63.2	78.0	84.0	88.7	91.8	91.1	82.2	71.4	58.9	51.9			
	0.4% MCWB	45.8	46.7	50.9	59.1	64.5	69.6	69.6	69.8	65.4	61.9	53.2	48.9			
	2% DB	46.6	49.0	58.1	72.0	79.4	83.9	87.1	85.3	76.4	66.5	55.1	48.4			
	2% MCWB	43.8	44.9	48.0	56.6	63.1	67.0	68.9	68.9	63.1	58.5	51.2	45.7			
	5% DB	43.5	46.0	53.8	66.9	75.4	79.9	83.2	81.6	72.5	62.5	51.9	45.8			
	5% MCWB	41.1	42.8	46.3	53.8	61.1	65.0	67.6	67.1	61.5	56.3	48.8	43.7			
	10% DB	40.8	42.9	49.7	62.5	71.4	76.4	79.7	78.1	68.7	59.1	49.6	42.8			
	10% MCWB	38.8	40.1	44.3	51.4	59.1	63.6	65.9	65.0	59.9	54.7	47.1	41.0			
	0.4% WB	46.9	47.7	52.5	60.5	67.2	71.4	72.9	72.7	67.9	62.7	54.5	49.9			
	0.4% MCDB	49.0	50.5	60.8	74.0	79.4	83.8	85.4	84.6	76.6	68.8	57.7	51.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	43.7	45.5	49.5	57.9	64.8	69.2	71.1	70.5	65.0	59.5	51.8	46.0			
	2% MCDB	45.9	48.4	56.3	70.1	75.9	79.9	82.7	81.6	73.5	64.6	54.3	47.6			
	5% WB	41.3	42.9	47.0	55.1	62.7	67.0	69.5	68.6	62.9	57.1	49.3	43.4			
	5% MCDB	43.3	45.7	52.2	64.3	72.7	76.9	79.7	78.6	70.1	61.7	51.4	45.2			
	10% WB	38.7	40.1	44.5	52.7	60.4	65.0	67.8	66.9	60.8	54.8	47.1	41.0			
	10% MCDB	40.4	42.7	49.2	60.3	69.2	73.7	76.6	75.5	67.3	58.9	49.1	42.6			
(35) Mean Daily Temperature Range	MDBR	8.7	10.2	13.7	17.8	18.5	17.8	17.8	18.2	16.4	13.8	9.0	8.3			
	5% DB	10.1	13.8	20.9	25.3	24.3	23.5	23.2	23.4	22.1	19.3	12.9	10.1			
	5% WB	8.2	10.4	12.7	12.6	11.1	10.3	9.0	10.0	10.8	11.9	9.5	8.3			
	MCDBR	9.8	12.5	19.0	22.5	21.9	21.0	20.2	21.1	19.6	17.7	11.8	9.6			
	MCWBR	8.3	10.0	12.7	11.9	11.3	10.5	9.2	9.6	10.7	11.6	9.4	8.3			
	MCWBR	8.3	10.0	12.7	11.9	11.3	10.5	9.2	9.6	10.7	11.6	9.4	8.3			
(40) Clear-Sky Solar Irradiance	taub	0.322	0.351	0.385	0.430	0.416	0.411	0.442	0.435	0.396	0.373	0.361	0.318			
	taud	2.225	2.211	2.198	2.141	2.241	2.282	2.212	2.231	2.309	2.338	2.274	2.264			
	Ebn at Noon	206	236	252	254	266	267	256	251	246	226	190	187			
	Edh at Noon	22	30	36	43	41	40	42	39	32	26	21	18			
	RadAvg	220	450	835	1296	1643	1743	1624	1424	1009	563	236	159			
(45) All-Sky Solar Radiation	RadStd	43	75	113	176	168	152	197	117	153	89	23	23			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.24	N/A	N/A	N/A	N/A	N/A	-268	-412	+207	+67
(47) Regional (24 neighbors)		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-297	+140	+50

Nomenclature: See separate page